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NEW ASPECTS OF DETERMINATION OF POLYMER HETEROGENEITY BY 2-DIMENSIONAL ORTHOGONAL LIQUID CHROMATOGRAPHY AND MALDI-TOF- MS

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Abstract: Liquid Adsorption Chromatography at Critical Conditions (LACCC) of polymers in combination with Matrix-Assisted Laser Desorption / Ionization Time-Of-Flight Mass Spectrometry (MALDI-TOF-MS) and different types of detectors is a very convenient method for the characterization of polymer heterogeneities. A pathway for working out these methods is given. Additionally to the coupling of different chromatographic modes a device for post-column reaction detection is described and applied.

Examples of applications are given for aliphatic and aromatic polyesters. MALDI-TOF-MS, reaction detection and relations between „critical" solvent composition and structure of polyesters are discussed. Conclusions concerning the mechanism of thermal degradation of polylactides are shown.

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INTRODUCTION

Liquid Adsorption Chromatography at Critical Conditions has become a well established method in the characterization of polymer heterogeneity (Refs. 1, 2).
First results on LACCC applications and on two-dimensional High Pressure Liquid Chromatography (HPLC) of polymers, realized by coupling LACCC in the first and Size Exclusion Chromatography (SEC) in the second dimension were obtained on polyesters and polyethers (Refs. 3, 4). MALDI-TOF Mass Spectrometry is a powerful supplement to Liquid Chromatography particular in the molar mass range up to 10000 g/mol (Ref. 5).
In this contribution we present both the progress in these methods and the new applications.
To understand the above chromatographic methods, the behaviour of dissolved polymer molecules in the stationary phase during the chromatographic separation process at different modes is presented in Fig. 1. In the case of ideal SEC (A) interactions with the stationary phase do not take place, and in an entropy-controlled process the separation proceeds according to the hydrodynamic volume of solute molecules. The dependence of elution volume on molar mass (D) is contrary in Liquid Adsorption Chromatography (LAC) or in Interaction Chromatography (C). In LAC the separation process is dominated by enthalpic differences. By compensation of entropic and enthalpic differences it is possible to realize „critical" conditions of polymer adsorption (B). In this case, the repeating units of the polymer chain do not contribute to separation and the substance is eluted independently of its molar mass, and heterogeneities of polymer sample give reasons for new chromatographic peaks. It should be stressed, that the compensation of entropic and enthalpic differences makes it possible to measure very small differences in polymer structure. The „critical" conditions depend in a very sensitive manner on the composition of the mobile phase, temperature, the water content and other parameters. Furthermore problems with solubility and preferential solvatation have to be considered if using a mixed mobile phase.
We think that the potential of LACCC is very large and till now only a little number of applications were investigated. There are many difficulties with reproducibility from one column to another. In the case of SEC all effort is necessary to avoid

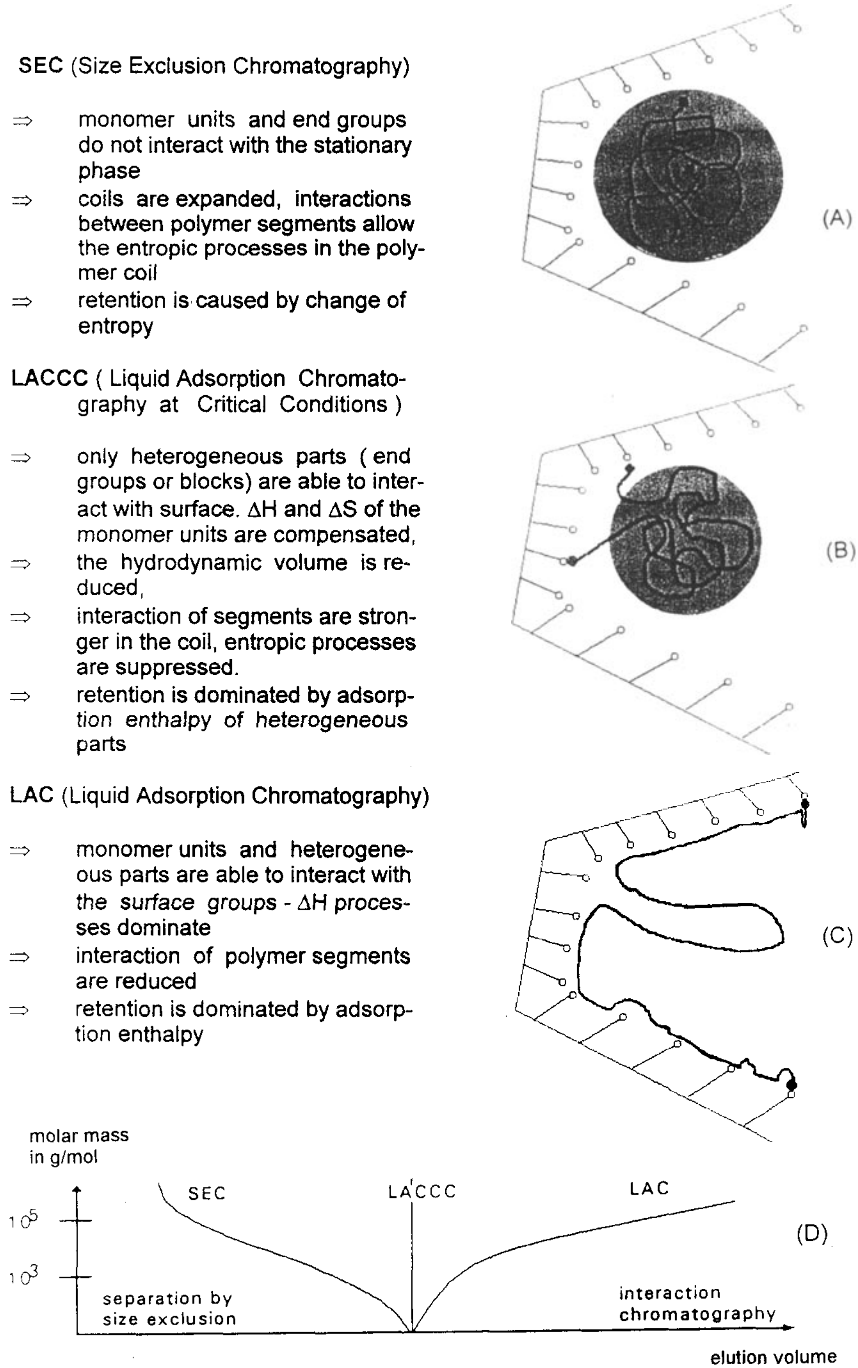


Fig.1. Three modes of Liquid Chromatography of polymers

interactions with stationary phase and the measurement of the elution volume has to be done as exactly as possible. In the case of LAC the aim is to realize great differences in retention, but the absolute elution volume is not so important. In the case of LACCC both the exact measurement of elution volume and a great selectivity for heterogeneous groups (groups with other functionality or another block in copolymers) are necessary for separation.

The choice of stationary phase is very important. Not only the pore size, which must be greater than the hydrodynamic volume of polymer molecules in the used mobile phase but the expected order of separation determines the choice between normal or reversed stationary phase. If the polarity of the heterogeneous part of macromolecule is greater than the polarity of the repeating unit, it is better to separate in normal phase mode. If the heterogeneous part is more hydrophobic the reversed phase separation should be used.

RESULTS AND DISCUSSION

Fig. 2 shows the pathway for the characterization of chemical heterogeneous polymers. To find out the „critical" conditions well characterized samples with different molar masses but equal chemical structure and optimal chromatographic phases are most important.

The necessary step after the chromatographic separation concerns the structural identification of peaks. Optimal in LACCC are detectors which suppress effects of mobile phase. Near the dead time in LAC displacement peaks or peaks of preferential solvatation appear. These are often changing their size (shape and position) from one day to the other. Therefore it is an advantage to use the Evaporative Light Scattering Detector (ELSD), the Matrix-Assisted Laser Desorption/ Ionization Time-Of-Flight Mass Spectrometry (MALDI-TOF-MS), or the FTIR-Detector. With ELSD it is possible to determine the mass content of chemically different structures. With MALDI-TOF-MS it is possible to get Molar Mass Distribution from one droplet of eluent from the peak maximum. In the case of ELSD we split the stream of eluent in front of ELSD in the relation of 20 to 80 percent, e.g., by different diameters of capillaries. After measurement of the offset it is possible to collect drops from the same run. For MALDI-TOF-MS the particular drops are dissolved and mixed with the optimized matrix. The advantage of this kind of mass spectrometry is not only its high sensitivity and the possibility to determine the

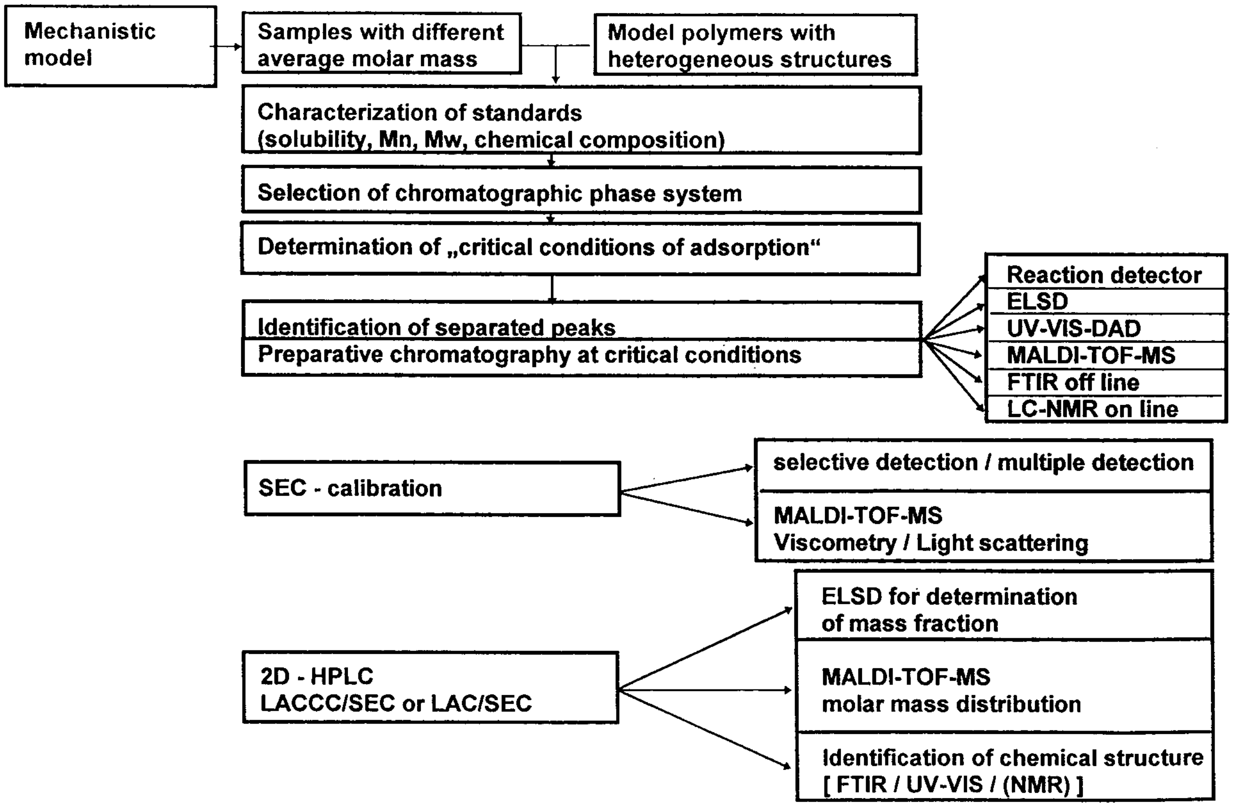


Fig.2. Pathway for characterization of chemically heterogeneous polymers

absolute molar masses of investigated substances but also the information about structure units and the structure of end groups by using the residual mass. On the one hand, it should be stressed, that some disadvantages exist in quantitative determination of polymer mixtures and influences of ionization, sample preparation, and the nature of the matrix - all affects the results obtained. On the other hand MALDI-TOF-MS is helpful for SEC-calibration.

Spectroscopic structure detection was realized by Diode-Array-Detector (DAD) or FTIR-instruments. Infrared structure detection is possible with FTIR-LC-Transform device (Lab Connections, USA). In this case all mobile phases were removed by heating or by ultrasonic nozzle. Another possibility presents post-column reaction detection which is selective for special functional groups (examples are presented in the following applications).

The scheme of two-dimensional chromatography with hyphenated detection is shown in Fig. 3.

In any case the first chromatographic mode is LACCC because it provides the separation independent of polymer molar mass. In dependence on the aim of investigation it is necessary to use the second chromatographic mode (SEC), or to identify the LACCC peaks by means of MALDI-TOF-MS, FTIR, DAD or by reaction detection.

Application of LACCC to polyesters

In our previous paper (Ref. 3) the identification of chromatographically separated peaks of polyesters consisting of adipic acid and 1,6-hexanediol (AA-HD) was done by chromatographic comparison with synthesized model substances. Now we present identification and characterization by MALDI-TOF Mass spectra of all fractions (Fig. 4). One droplet of every peak, nearly 0.5 µL was collected and the matrix (2,5-dihydroxybenzoic acid in ethanol) was added and mixed with the sample in the suitable ratio. Molar mass distribution of polyester can be calculated from the MALDI-TOF mass spectrum. The monomer unit and the structure of end groups from the residual mass can be determined by using the software of Kompact MALDI III equipment of Kratos/Shimadzu. All molecules are indicated as Na^+-adducts.

The influence of polyester structure on critical composition of eluent is shown in Table 1. The identified critical composition of eluent in dependence on polyester

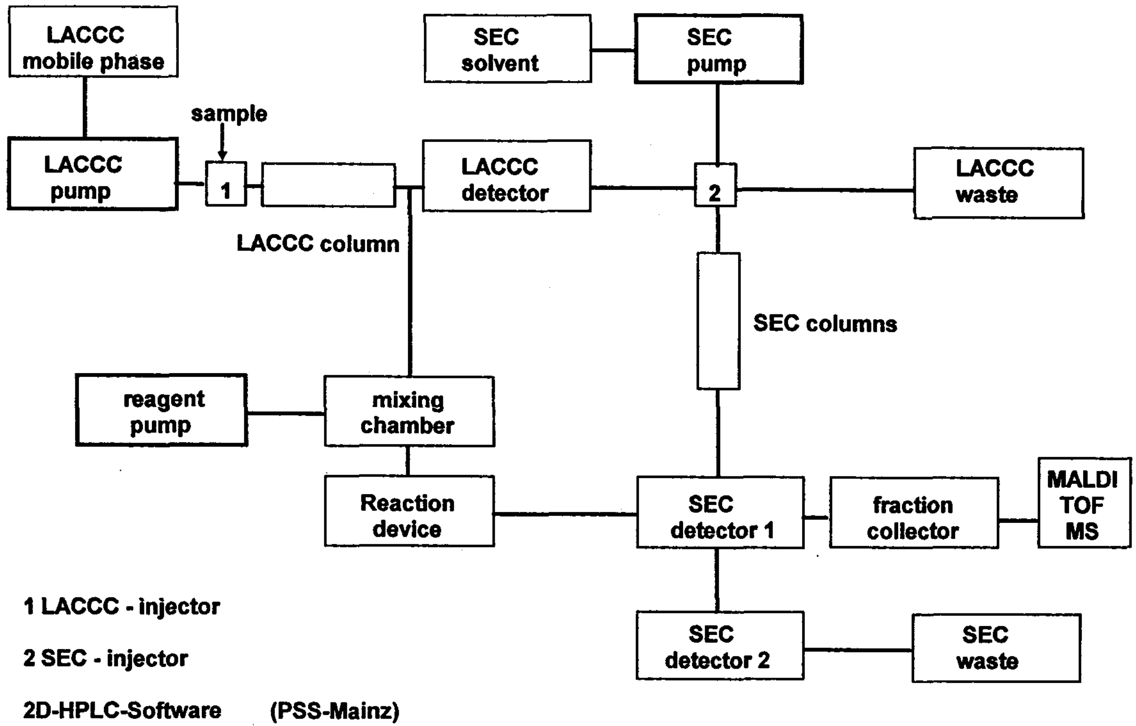


Fig.3. Scheme of two-dimensional chromatography with hyphenated detection

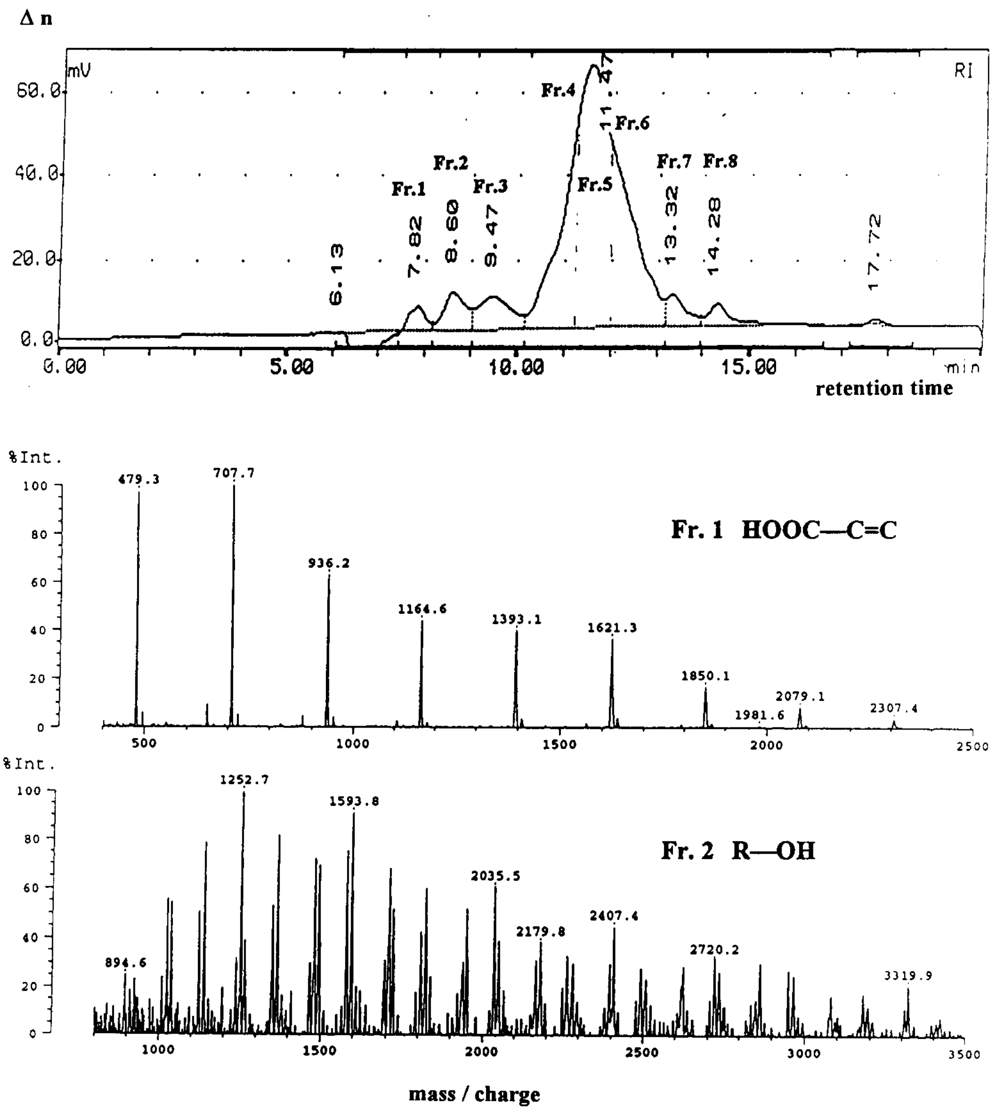


Fig.4a. LACCC (RI detector) of AA-HD polyester and MALDI-TOF-MS of fractions 1 and 2

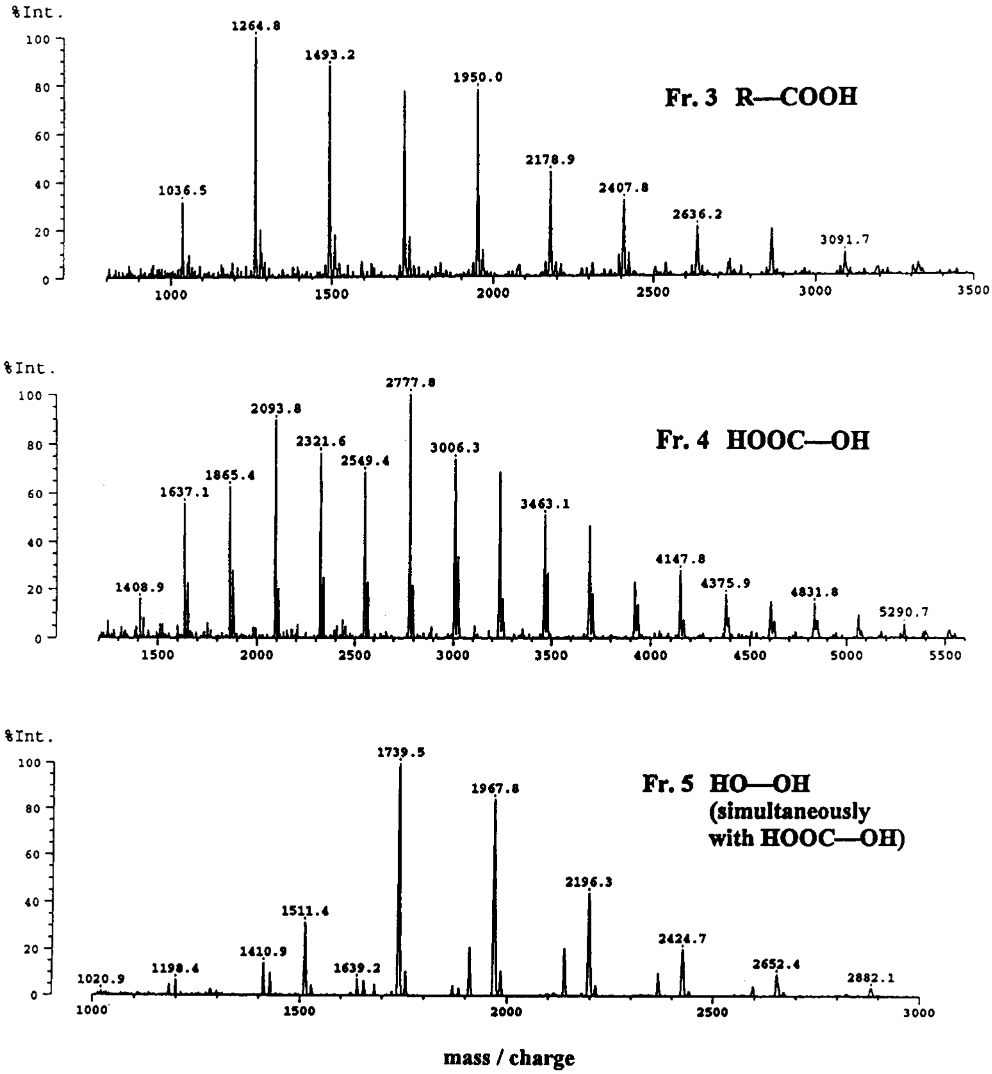


Fig.4b. MALDI-TOF-MS of polyester fractions 3-5

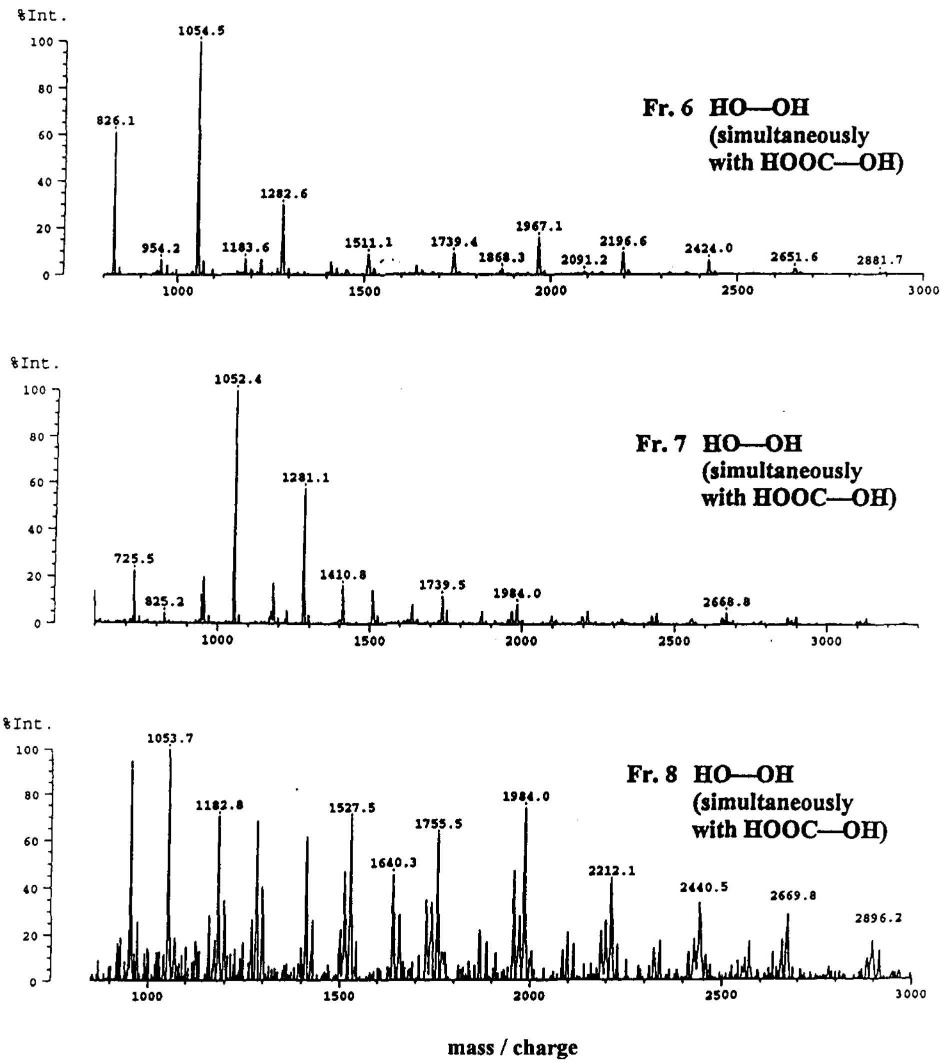


Fig.4c. MALDI-TOF-MS of polyester fractions 6-8

structure is presented in Table 1 too. The chain length of diol varied from 2 to 6. The shorter or more polar the diol the higher content of acetone must be used to reach critical solvent composition.

The critical solvent composition increases with the number of ethylene oxide units in the diol. In the case of phthalic acid polyesters with different diols which are only soluble in mixtures of tetrahydrofuran (THF).

Tab.1. Mobile phase composition at „critical" conditions of polyesters in dependence on structure at 35°C

Acid	**Diol**	**Mobile phase** (vol.-%)	**Stationary phase**
Adipic acid	1,6-Hexanediol	51 % Acetone / 49 % Hexane	Separon SGX 120 Å
Adipic acid	1,4-Butanediol	63 % Acetone / 37 % Hexane	Separon SGX 120 Å
Adipic acid	1,3-Butanediol	62 % Acetone / 38 % Hexane	Separon SGX 120 Å
Adipic acid	1,2-Propanediol	64 % Acetone / 36 % Hexane	Separon SGX 120 Å
Adipic acid	1,3-Propanediol	60 % Acetone / 40 % Hexane	Separon SGX 120 Å
Adipic acid	1,2-Ethanediol	70 % Acetone / 30 % Hexane	Separon SGX 120 Å
Adipic acid	Neopentyl glycol	53 % Acetone / 47 % Hexane	Separon SGX 120 Å
Adipic acid	Diethylene glycol	40 % Acetone / 60 % Hexane	Separon SGX 120 Å
Adipic acid	Dipropylene glycol	60 % Acetone / 40 % Hexane	Separon SGX 120 Å
Phthalic acid	Ethylene glycol	66 % THF / 34 % Heptane	Spherisorb SAX 120Å
Phthalic acid	Diethylene glycol	72.5% THF / 27.5% Heptane	Spherisorb SAX 120Å
Phthalic acid	Triethylene glycol	100 % THF	Separon SGX 120 Å

Fig. 5 presents the principle of post-column reaction detection device. A reaction solution is pumped with constant flow into a mixing chamber by means of an additional chromatographic system. A selective reaction with specific functional groups produces changing of optical or electrochemical properties of the fraction after mixing with the eluent of LACCC. Most important is that the reaction has to be fast. If the reaction is not fast enough, a reaction coil must be used.

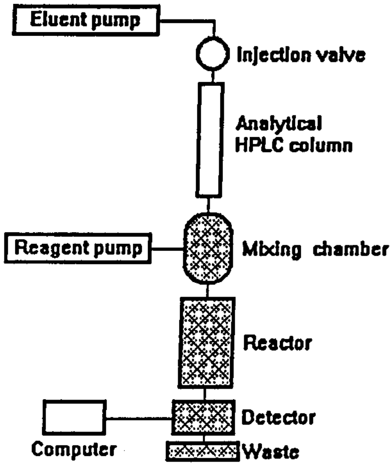


Fig.5. Scheme of post-column reaction detection

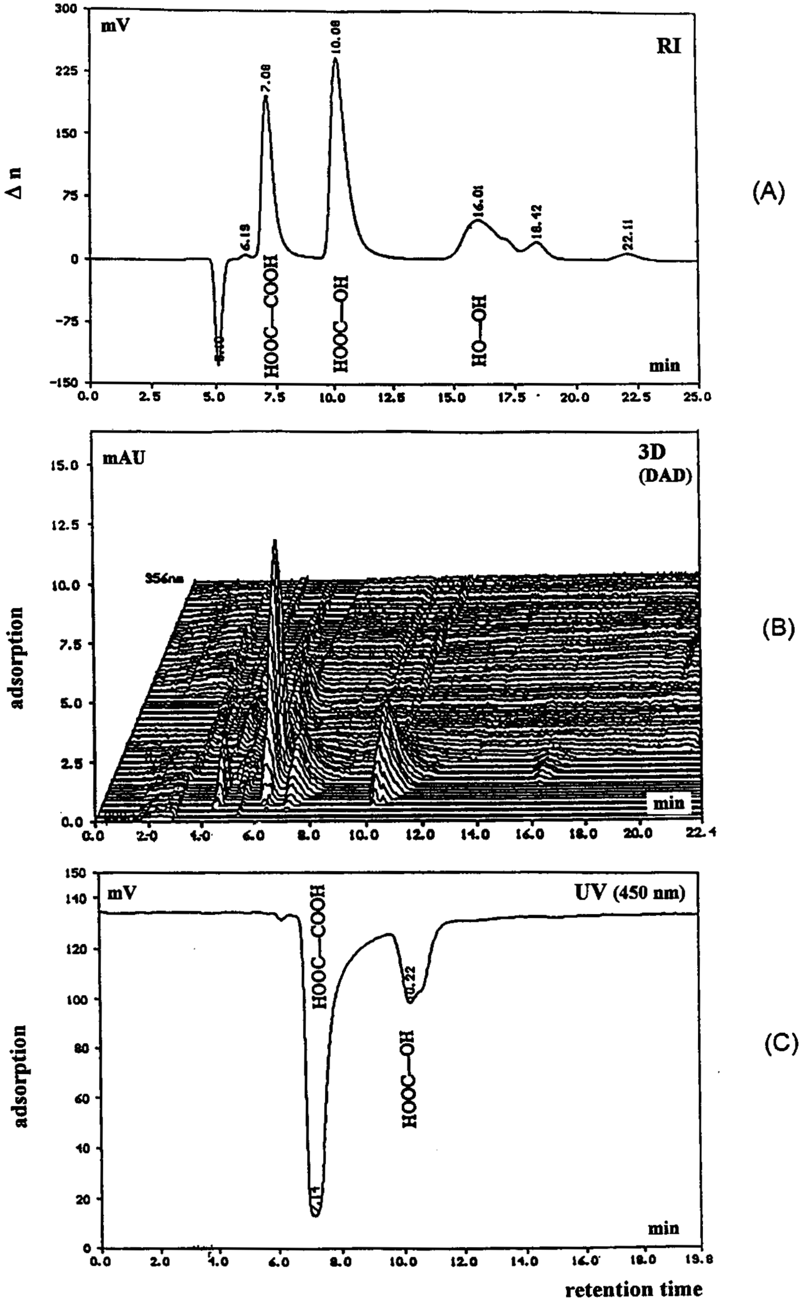


Fig.6. Post-column reaction detection of carboxylic end groups

Advantages of this method are: contrary to derivatization before separation, artefacts do not influence the detection, substances were separated in original form, reaction has not to be complete but reproducible.

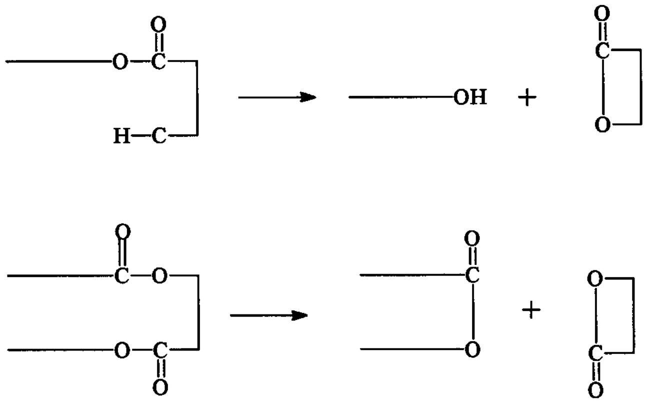
Very often, however, eluent is not the optimal medium for reaction, excess of reagent can disturb the signal of the sample, band broadening can reduce chromatographic resolution, and the application is limited by low velocity of some reactions. In this example a fast protolytic reaction of the sodium salt of the weak acid o-nitrophenol (10 mg sodium nitrophenolate in 100 ml of the critical mobile phase) with the carboxylic group of eluted components was applied. The disappearance of salt anion resulting in a negative peak was measured by spectroscopy. The LACCC chromatograms of adipic acid/1,6-hexanediol polyester with different carboxylic end groups are shown in Fig.6. the detection with refractive index detector is presented in chromatogram (A), while the detection with UV-DAD-detector (200-356 nm) is shown in chromatogram (B) and detection with post-column reaction by spectroscopy at 450 nm in chromatogram (C). OH-terminated species are not indicated in (C) and the intensities of peaks correlate with the content of carboxylic groups.

Polylactides

The aim was to investigate the mechanism which is behind thermal degradation of polylactides applying the above mentioned chromatographic and spectroscopic methods (Ref. 6, 7).

All experiments were done with a thermally treated and untreated oligo-L-lactide as model substances. The oligolactide had initial molar mass lower than 1000 g/mol. These oligolactides were used because they have higher end group content and better ability of ionization and detection in the MALDI-TOF-MS analysis. The oligomers were heated in closed glass tubes in nitrogen atmosphere at 200°C in a time program.

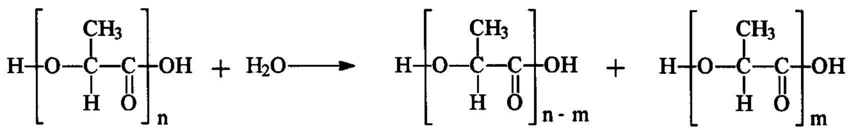
Mechanisms of the thermal degradation of polylactide were proposed in the literature (Refs. 8 - 11). They include the intramolecular transesterification **1** by formation of cyclic oligomers and linear oligolactide, the intermolecular transesterification **2** and the ester hydrolysis **3**, by formation of linear polyesters and the pyrolytic elimination **4**, connected with the formation of acrylic and carboxylic end groups.



1

2

R : polylactide segments



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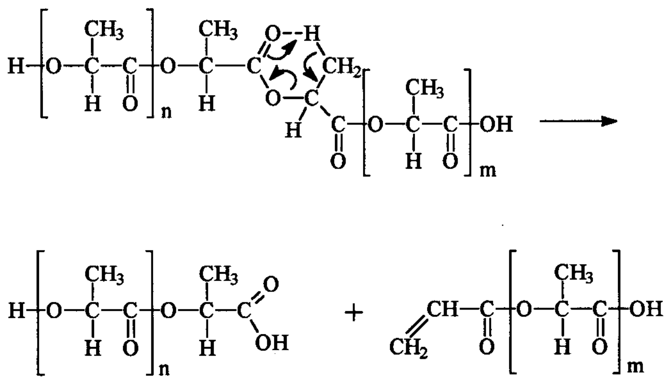


4

Fig. 7 shows the molar mass dependence of elution volumes of poly-L-lactides at different solvent compositions. Critical solvent composition of eluent is reached, when elution volume is independent of molar mass. We used 1,4-dioxane as a good solvent and hexane was added successively as a thermodynamically poor solvent.

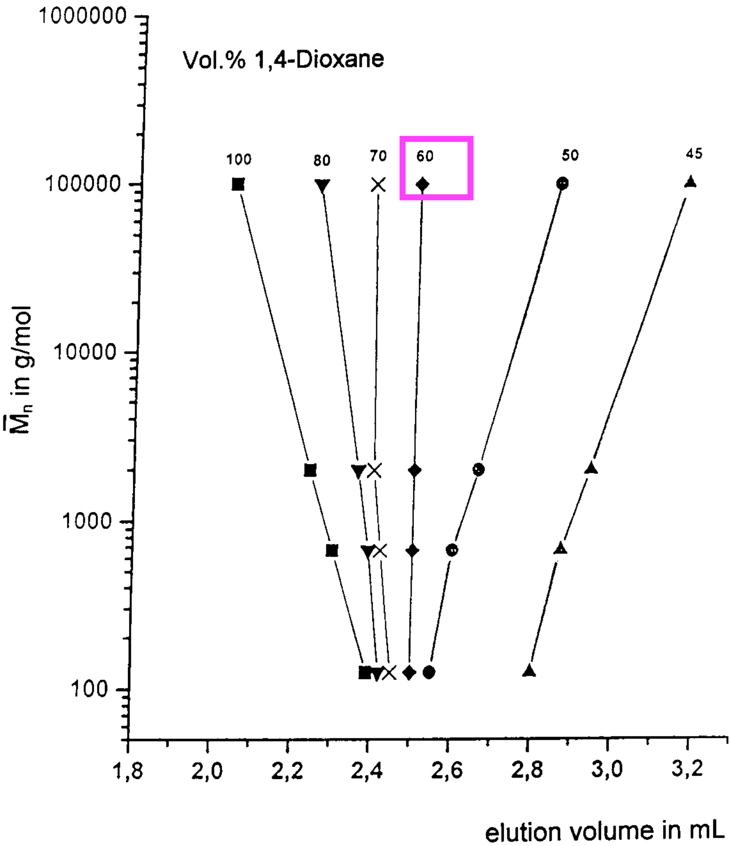
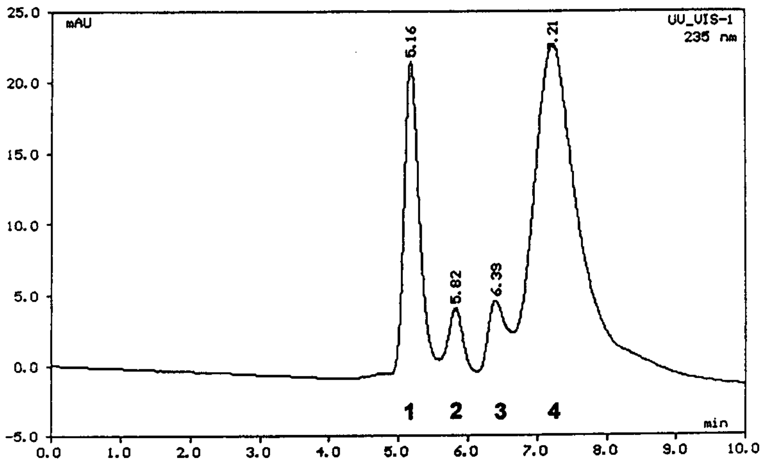
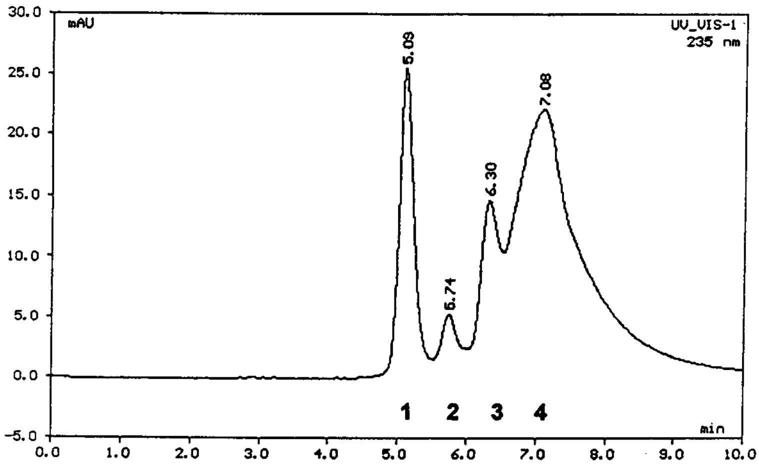


Fig.7. Determination of critical solvent composition of poly-L-lactide in n-hexane, Nucleosil 100 (250 x 4 mm inner diameter), 30°C



(A)



(B)

Fig.8. LACCC of oligolactide (A) and thermally treated oligolactide (B) at 235 nm

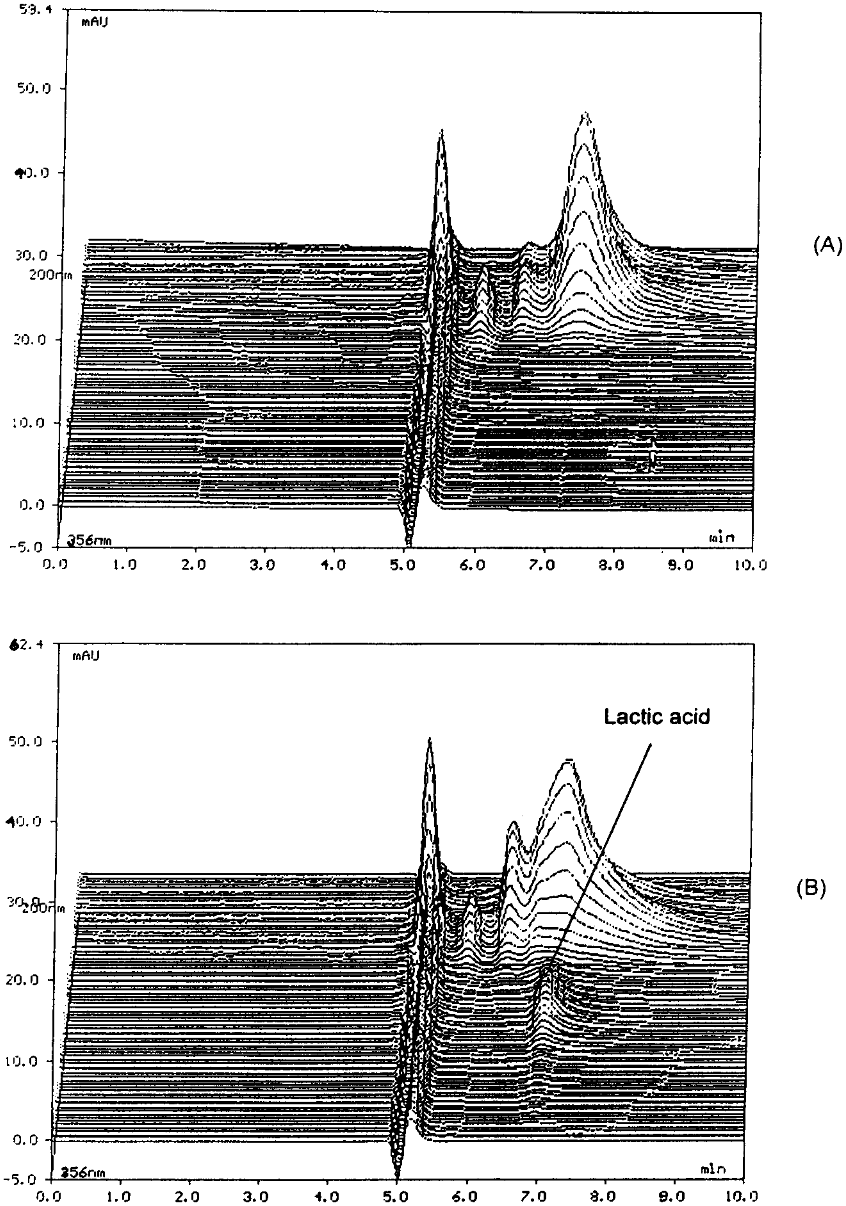


Fig.9. LACCC of oligolactide (A) and thermal treated oligolactide (B), measured with a Diode-Array-Detector

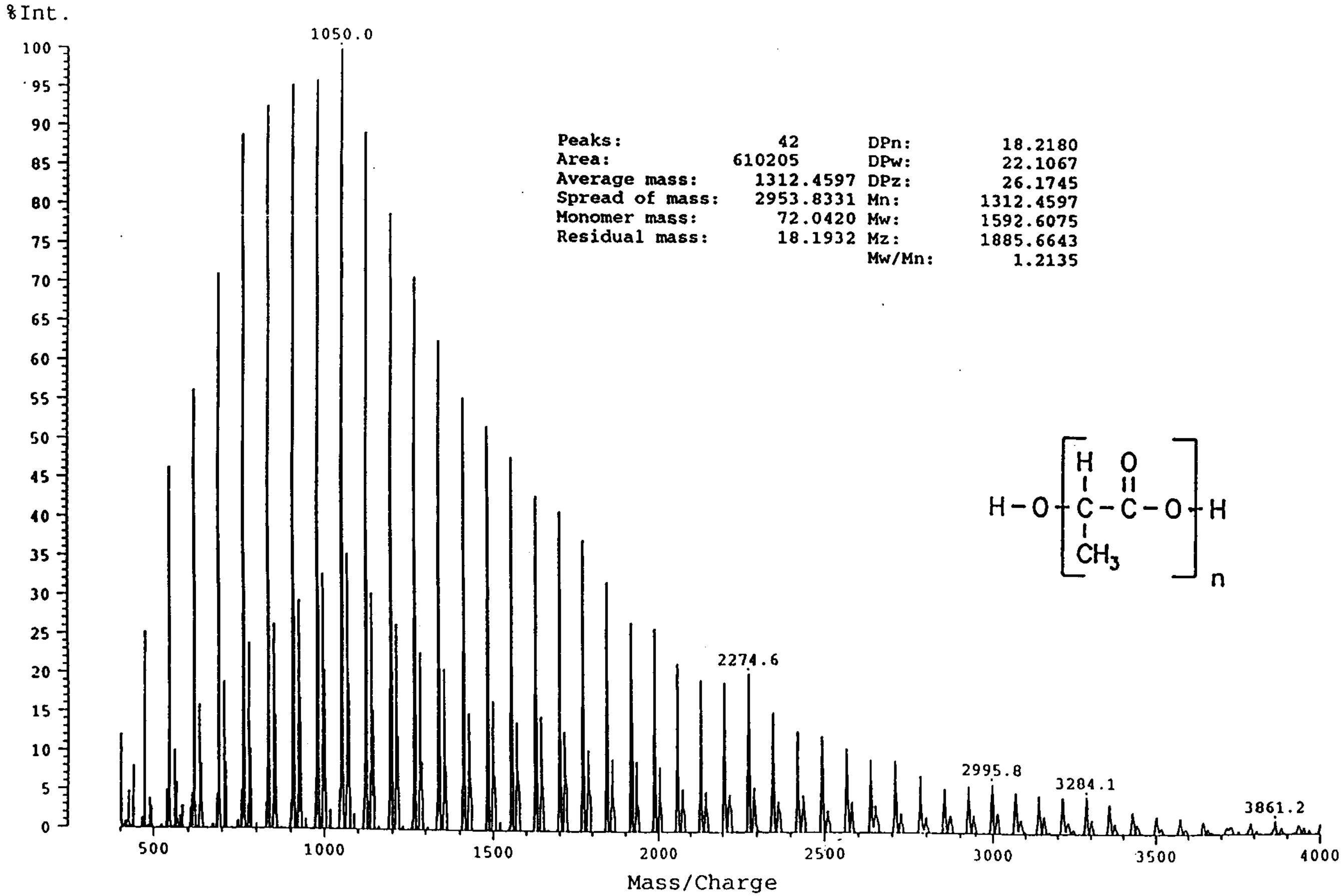


Fig.10a. MALDI-TOF-MS of peak 3

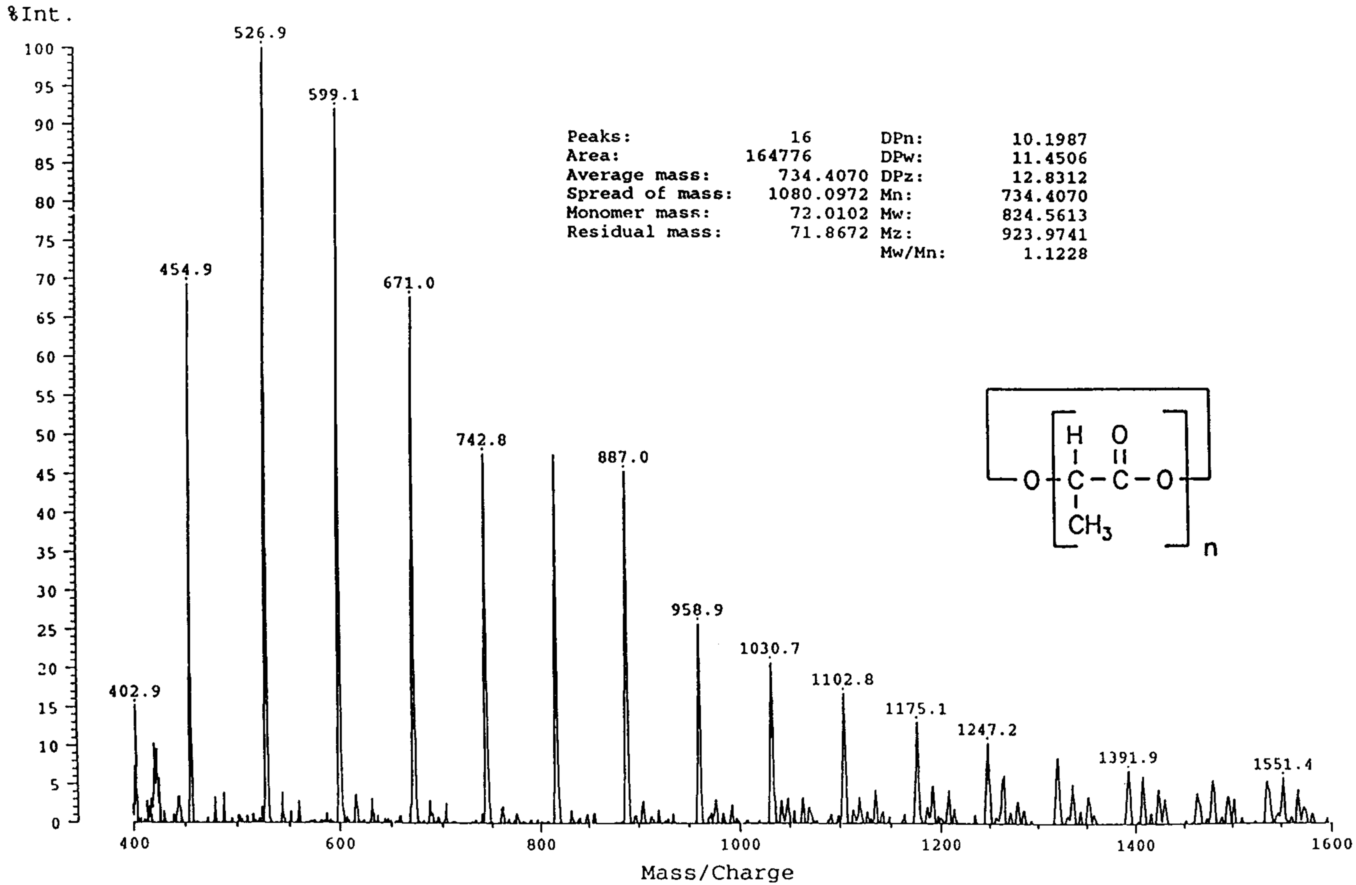


Fig.10b. MALDI-TOF-MS of peak 4

Fig. 8 and Fig. 9 show the obtained critical chromatograms at 235 nm and the three-dimensional wavelength plot from polylactide (A) in comparison to the thermally treated sample (B). Four characteristic peaks were observed in both cases, but with different intensities. Furthermore the thermally treated sample shows a new peak at a red- shifted wavelength at the same retention time as the peak 4 in the UV-DAD, Fig. 9, (B). This peak was identified as lactic acid by chromatographic comparison. The peak identification in the chromatograms of Fig. 8 was done by MALDI-TOF-MS, post-column reaction detection and by chromatographic comparison. Peak **1** is a displacement peak of mobile phase, proved by injection of the critical mixture. Peak **3** and peak **4** were identified as cyclic oligomers and linear polyesters, respectively, by MALDI-TOF-MS, Fig. 10. The residual mass of cycles (0 g/mol) is the same as from the structure unit, but acrylic acid end group have the same mass, too.

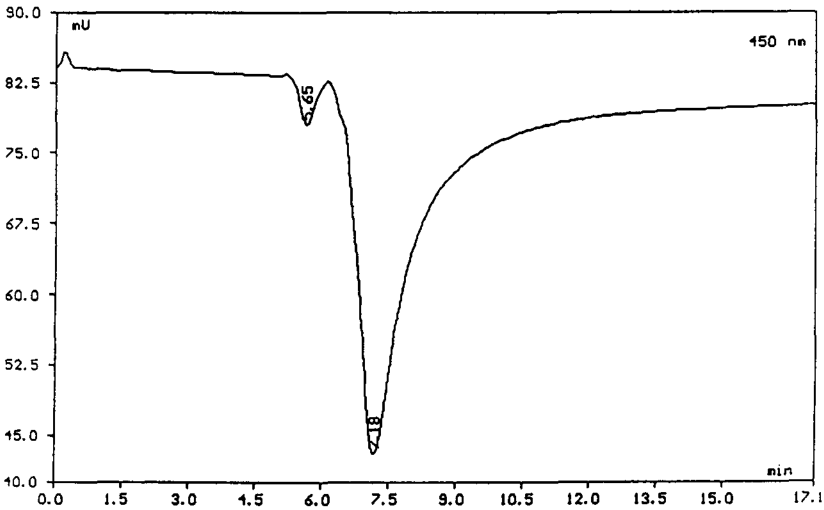


Fig.11. Identification of peak **2** and peak **4** of Fig. 8 by the reaction detection at 450 nm

As shown in Fig. 11 only the peak **2** and peak **4** of the chromatogram in Fig. 8 give a signal with reaction detection. This proves that only these peaks have free carboxylic end groups. Finally peak **2** could be identified as a polymer with acrylic end groups by chromatographic comparison.

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